

Integrated Campus Medical Facility & Patient Management System

A Full-Stack MERN Application for Smart Campus Healthcare Administration



Summary / Abstract

The Integrated Campus Medical Facility & Patient Management System is a full-stack MERN web application that digitizes campus healthcare services. It replaces manual dispensary workflows with a secure role-based platform for managing appointments, health records, prescriptions, medicine inventory, emergency alerts, and notifications, improving accessibility, transparency, security, and operational efficiency.

Problem Solved

- Paper-based medical records are hard to search, update, audit, and protect.
- Manual medicine stock tracking can cause shortage, expiry, and mismatch issues.
- Students do not get quick visibility of appointments, prescriptions, and health history.
- Emergency communication through notice boards or informal chats is slow and unreliable.
- Dispensary staff spend extra time on repeated administrative work instead of patient care.

Objective / Key Goals

To develop a secure, scalable, and role-based healthcare platform that modernizes campus dispensary operations through centralized digital records, automated inventory tracking, and structured communication.

Key Goals:

- Separate dashboards for Student, Doctor, Staff, and Admin.
- Digitize health records, appointments, prescriptions, medicines, and notifications.
- Use secure authentication and role-based authorization.
- Provide faster emergency alert delivery and real-time updates.
- Build a maintainable architecture that can be extended for mobile app, analytics, and telemedicine.

Student

Doctor

Staff

Admin

Method / Approach

The system follows a MERN-based three-tier architecture for better scalability, maintainability, and security.

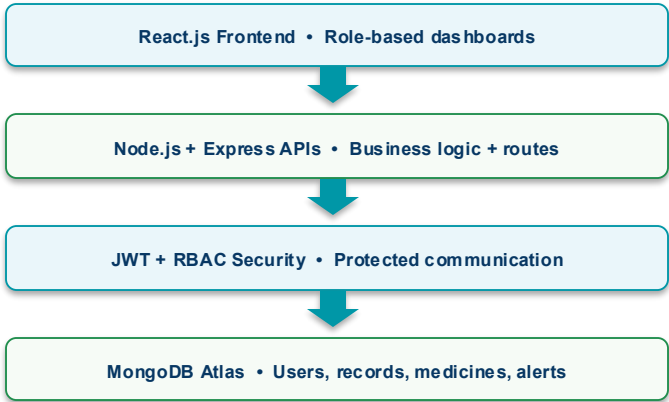
Frontend Layer:

React.js, Vite, Tailwind CSS, React Router, Context API, and Axios provide responsive role-based interfaces.

Backend Layer:

Node.js and Express.js expose REST APIs for auth, facilities, medicines, hospitals, notifications, gallery, history, appointments, emergency alerts, and admin operations.

System Architecture



Database Layer:

MongoDB Atlas stores users, health records, appointments, medicines, notifications, gallery data, and admin-related records. Mongoose handles connectivity and schema management.

Security Approach



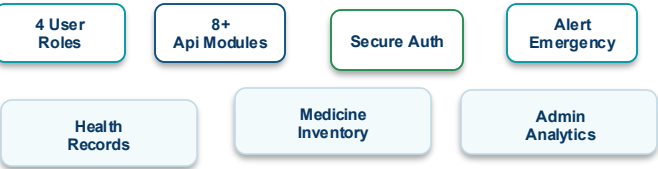
Results / Findings

The project delivers a production-ready, role-based campus healthcare platform with secure login, OTP verification, digital health records, prescriptions, inventory tracking, appointments, nearby hospitals, emergency alerts, notifications, admin analytics, and gallery management.

Its modular REST API backend centralizes all dispensary workflows into one secure, maintainable, scalable, and easy-to-extend digital system.

Key Outcomes

- Complete MERN full-stack healthcare platform
- Secure multi-role access for Student, Staff, Doctor, Admin
- Digitized records, prescriptions, inventory, alerts and notifications
- Reduced paperwork and improved operational transparency
- Scalable architecture for future deployment



Conclusion / Significance

1.ICMFMS uses modern full-stack engineering to digitize campus healthcare through role-based dashboards, secure authentication, medical records, inventory tracking, alerts, notifications, and admin visibility.

2.It improves speed, reliability, transparency, and accessibility while remaining practical, scalable, secure, and ready for future AI, mobile, telemedicine, analytics, and cloud extensions.